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Chapter 1

Viscoelastic Fluid Mechanics

In this chapter, the continuum mechanics framework governing the flow of incompressible viscoelastic fluids is rigorously established. The primary objective is to derive the complete set of differential operators and constitutive equations that define the physical domain and formulate the residuals for the Physics-Informed Neural Network (PINN) architecture. We begin from the universal conservation principles of mass and linear momentum in three dimensions, and subsequently introduce the physical and geometric assumptions characterizing micro-scale polymer flows in enclosed geometries. The Oldroyd-B constitutive model is then formulated in detail, emphasizing the requirement of material frame-indifference via the Upper-Convected Time Derivative (UCTD) and discussing the rheological implications of extensional kinematics. A dedicated non-dimensionalization procedure, based on viscous stress scaling, is presented to ensure well-conditioned gradients within numerical optimization. Finally, we formalize the four-roll mill benchmark, describing its hyperbolic stagnation-point kinematics, boundary conditions, and the mathematical treatment of the pressure gauge freedom.

1.1 General Conservation Laws

From a continuum mechanics perspective, the spatial and temporal motion of a fluid continuum occupying an open domain $\Omega \subset \mathbb{R}^3$ is governed by the universal conservation laws of mass and linear momentum [1].

1.1.1 Principle of Mass Conservation

The differential statement of mass conservation for a fluid with local density $\rho(\mathbf{x}, t)$ and velocity vector $\mathbf{u}(\mathbf{x}, t)$ is expressed through the continuity equation:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot (\rho \mathbf{u}) = 0. \quad (1.1)$$

Equation (1.1) dictates that the instantaneous rate of mass accumulation within any arbitrary fixed control volume must be strictly balanced by the net convective mass flux through its bounding control surface.

1.1.2 Conservation of Linear Momentum

Applying Newton's second law to an infinitesimal fluid parcel yields the Cauchy momentum equation in Eulerian coordinates:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = \nabla \cdot \boldsymbol{\sigma} + \rho \mathbf{g}, \quad (1.2)$$

where $\mathbf{g}$ denotes the acceleration due to external body forces (gravity), and $\boldsymbol{\sigma}$ is the symmetric second-order Cauchy stress tensor. The left-hand side of Equation (1.2) represents the material (substantial) acceleration $D\mathbf{u}/Dt$, which combines the local unsteady rate of change $\partial \mathbf{u}/\partial t$ with the non-linear convective acceleration term $\mathbf{u} \cdot \nabla \mathbf{u}$ stemming from spatial velocity gradients.

To distinguish between isotropic pressure forces and internal shear/normal

deviatoric stresses, the Cauchy stress tensor is decomposed as:

$$\boldsymbol{\sigma} = -p\mathbf{I} + \mathbf{T}, \quad (1.3)$$

where p is the isotropic thermodynamic pressure, $\mathbf{I}$ is the identity tensor, and $\mathbf{T}$ is the extra-stress (or deviatoric stress) tensor. The extra-stress tensor encapsulates all viscous dissipation and elastic energy storage mechanisms originating from the deformation of the fluid microstructure. Substituting Equation (1.3) into Equation (1.2) gives:

$$\rho \left(\frac{\partial \mathbf{u}}{\partial t} + \mathbf{u} \cdot \nabla \mathbf{u} \right) = -\nabla p + \nabla \cdot \mathbf{T} + \rho \mathbf{g}. \quad (1.4)$$

1.2 Physical Assumptions and Governing System Reduction

The full three-dimensional, unsteady equations of motion are intractable without specific constitutive closures and scale-appropriate reductions. For the analysis of polymer solutions within the four-roll mill geometry, we introduce the following physical assumptions:

1. **Single-phase continuum:** The polymer solution is treated as an isotropic, single-phase, homogeneous continuum. Individual polymer macromolecules and solvent molecules are not resolved discretely; rather, their collective thermodynamic and rheological effects are captured through effective macroscopic transport properties.
2. **Incompressibility and solenoidal velocity field:** Under the operating pressures and low flow velocities characteristic of liquid polymer systems, density variations are negligible ($\rho = \text{const}$). The continuity equation (1.1)

reduces to the solenoidal constraint:

$$\nabla \cdot \mathbf{u} = 0. \quad (1.5)$$

A solenoidal velocity field implies that the velocity vector is divergence-free throughout the entire flow domain. Physically, this means that every material parcel preserves its volume identically during deformation (isochoric motion), and the flow possesses neither volumetric mass sources nor sinks.

3. **Isothermal conditions:** Viscous dissipation in the present low-shear regime produces negligible internal temperature rises. Thermal uniformity is maintained, decoupling the energy conservation equation from momentum and allowing fluid properties (viscosities, relaxation time) to be treated as spatially invariant constants.
4. **Negligible gravitational body forces:** In micro- and milli-fluidic geometries with characteristic length scales on the order of millimeters to centimeters ($L \sim 10^{-2}$ m) and viscous working fluids ($\mu \sim 0.1 - 1$ Pa · s), gravitational body forces are completely overshadowed by viscous and viscoelastic stress gradients. This is confirmed by evaluating the ratio of hydrostatic pressure gradients ρg to characteristic viscous shear forces $\mu_0 U/L^2$:

$$\frac{\rho g L^2}{\mu_0 U} \ll 1. \quad (1.6)$$

Consequently, body force contributions are neglected by setting $\mathbf{g} = \mathbf{0}$.

5. **Two-dimensional steady-state kinematics:** The geometry of the four-roll mill features rollers with a high aspect ratio along the axial dimension (z -axis). In the transverse mid-plane of the device, edge effects from the top

and bottom confinement are negligible, rendering the flow strictly planar:

$$\mathbf{u} = \begin{pmatrix} u(x, y) \\ v(x, y) \\ 0 \end{pmatrix}, \quad \frac{\partial(\cdot)}{\partial z} = 0. \quad (1.7)$$

Furthermore, once the boundary roller rotations are brought to constant angular speeds, the fluid reaches an established laminar regime where all local temporal derivatives vanish identically ($\partial(\cdot)/\partial t = \mathbf{0}$).

6. Deviatoric stress decomposition: For dilute and semi-dilute polymer solutions, the total extra-stress tensor $\mathbf{T}$ is partitioned into a purely viscous Newtonian solvent contribution and an elastic polymeric contribution $\boldsymbol{\tau}$ [1, 2]:

$$\mathbf{T} = 2\mu_s \mathbf{D} + \boldsymbol{\tau}, \quad (1.8)$$

where μ_s represents the dynamic viscosity of the Newtonian solvent, and $\mathbf{D}$ denotes the symmetric rate-of-strain tensor:

$$\mathbf{D} = \frac{1}{2} [\nabla \mathbf{u} + (\nabla \mathbf{u})^T]. \quad (1.9)$$

This two-fluid stress split reflects the fundamental rheological physics of polymer solutions: the Newtonian solvent responds instantaneously to shear through viscous dissipation ($2\mu_s \mathbf{D}$), whereas the flexible polymer coils experience flow-induced stretching and orientation, giving rise to an elastic restoring stress $\boldsymbol{\tau}$ governed by macromolecular relaxation.

Substituting the stress decomposition (1.8) and the physical simplifications into Equation (1.4) yields the reduced steady two-dimensional momentum equation:

$$\rho(\mathbf{u} \cdot \nabla \mathbf{u}) = -\nabla p + \mu_s \nabla^2 \mathbf{u} + \nabla \cdot \boldsymbol{\tau}. \quad (1.10)$$

Here, the Laplacian term $\mu_s \nabla^2 \mathbf{u}$ arises directly from $\nabla \cdot (2\mu_s \mathbf{D}) = \mu_s \nabla^2 \mathbf{u}$ by virtue of incompressibility ($\nabla \cdot \mathbf{u} = 0$). Equation (1.10) represents the Navier–Stokes momentum balance coupled directly to the divergence of the polymeric extra-stress tensor, $\nabla \cdot \boldsymbol{\tau}$.

1.3 Viscoelastic Constitutive Modeling: The Oldroyd-B Fluid

The momentum equation (1.10) contains six unknown scalar fields in 2D ($u, v, p, \tau_{xx}, \tau_{xy}, \tau_{yy}$), but provides only three equations (continuity and two momentum components). To mathematically close the boundary value problem, a constitutive equation governing the evolution of the polymeric extra-stress tensor $\boldsymbol{\tau}$ is required.

1.3.1 Microscopic Formulation and Dimensional Constitutive Law

In this work, we adopt the classical Oldroyd-B constitutive model [3]. From a kinetic theory perspective, the Oldroyd-B model represents polymer molecules as dilute, non-interacting Hookean dumbbells immersed in a Newtonian continuous medium. Each dumbbell comprises two Brownian beads connected by a linear, infinitely extensible entropic spring [1]. The beads experience viscous drag forces from the solvent, while the internal spring exerts an entropic restoring force proportional to the end-to-end vector of the dumbbell.

Under these microstructural assumptions, the polymeric stress tensor $\boldsymbol{\tau}$ is governed by the differential constitutive equation:

$$\lambda \overset{\nabla}{\boldsymbol{\tau}} + \boldsymbol{\tau} = 2\mu_p \mathbf{D}, \quad (1.11)$$

where λ is the characteristic polymer relaxation time, μ_p is the polymeric viscosity contribution, and $\overset{\nabla}{\boldsymbol{\tau}}$ denotes the objective time derivative. The total zero-shear viscosity of the solution is given by the sum of both phases:

$$\mu_0 = \mu_s + \mu_p. \quad (1.12)$$

1.3.2 Material Frame-Indifference and the Upper-Convected Time Derivative

In non-Newtonian fluid mechanics, constitutive equations cannot employ the standard partial time derivative $\partial\boldsymbol{\tau}/\partial t$ nor the substantial derivative $D\boldsymbol{\tau}/Dt$, as these operators are not invariant under time-dependent Euclidean transformations of the observer frame (principle of material frame-indifference or objectivity) [1, 4].

To account for the continuous translation, rotation, and stretching of fluid elements containing oriented polymer chains, the rate of change of $\boldsymbol{\tau}$ must be evaluated using the *Upper-Convected Time Derivative* (UCTD), denoted by $\overset{\nabla}{\boldsymbol{\tau}}$:

$$\overset{\nabla}{\boldsymbol{\tau}} \equiv \frac{\partial\boldsymbol{\tau}}{\partial t} + \mathbf{u} \cdot \nabla\boldsymbol{\tau} - (\nabla\mathbf{u}) \cdot \boldsymbol{\tau} - \boldsymbol{\tau} \cdot (\nabla\mathbf{u})^T. \quad (1.13)$$

In Equation (1.13), each term embodies a clear physical mechanism:

$\partial\boldsymbol{\tau}/\partial t$ is the local rate of stress change at a fixed spatial coordinate;

$\mathbf{u} \cdot \nabla\boldsymbol{\tau}$ represents the convective advection of stress carried along the flow streamlines;

$-(\nabla\mathbf{u}) \cdot \boldsymbol{\tau} - \boldsymbol{\tau} \cdot (\nabla\mathbf{u})^T$ are the non-linear tensorial stretching and rotation terms. They quantify how local velocity gradients deform and rotate the macromolecular dumbbell conformations, aligning the polymer stress with the principal axes of extension.

For the steady-state regime investigated in this research ($\partial\boldsymbol{\tau}/\partial t = \mathbf{0}$), the UCTD reduces strictly to the convective and deformation-induced stretching contributions.

1.3.3 Rheological Behavior and the High-Weissenberg Challenge

The Oldroyd-B model is widely recognized as the fundamental benchmark for elastic Boger fluids—polymer solutions formulated to exhibit significant elasticity while maintaining an essentially constant shear viscosity over a broad range of shear rates ($\eta(\dot{\gamma}) = \mu_0$).

However, in strong extensional kinematics (such as those generated along the axes of a four-roll mill), the Oldroyd-B model reveals an intrinsic mathematical singularity. Because Hookean dumbbells possess an infinitely extensible spring law, the steady extensional viscosity η_E predicted in a planar extensional flow field ($u = \dot{\epsilon}x, v = -\dot{\epsilon}y$) takes the analytical form:

$$\eta_E(\dot{\epsilon}) = 4\mu_s + \frac{2\mu_p}{(1 - 2\lambda\dot{\epsilon})(1 + 2\lambda\dot{\epsilon})}. \quad (1.14)$$

As the dimensionless extensional rate approaches the critical threshold $2\lambda\dot{\epsilon} \rightarrow 1$ (corresponding to an extensional Weissenberg number $Wi \rightarrow 0.5$), the polymeric stress diverges asymptotically to infinity ($\eta_E \rightarrow \infty$).

In numerical computations and machine learning architectures alike, this steep stress amplification produces extremely sharp spatial boundary layers along outflow streams, giving rise to the classical *High-Weissenberg Number Problem* (HWNP). Capturing these localized, steep stress gradients without inducing numerical instabilities or gradient explosion is one of the primary motivations for adopting physics-informed neural network approaches.

1.4 Nondimensionalization and Governing Equations

Nondimensionalization scales variables to order unity, improving numerical conditioning and highlighting the non-dimensional parameters controlling the flow regime [4]. In highly viscous and low-Reynolds viscoelastic flows (such as microfluidics and polymer processing in the four-roll mill), choosing a *viscous stress scale* rather than an inertial scale prevents loss of precision and gradient stiffness in the neural network optimization.

1.4.1 Reference Scales and Dimensionless Parameters

Selecting a reference length H_{ref} (e.g., roller radius), a reference velocity U_{ref} (roller surface speed), and a constant reference viscosity scale η_0 (e.g., characteristic total viscosity), we define the dimensionless coordinates, velocities, pressure, and extra-stress tensor:

$$\mathbf{x}^* = \frac{\mathbf{x}}{H_{\text{ref}}}, \quad \mathbf{u}^* = \frac{\mathbf{u}}{U_{\text{ref}}}, \quad p^* = \frac{p}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}, \quad \boldsymbol{\tau}^* = \frac{\boldsymbol{\tau}}{\frac{\eta_0 U_{\text{ref}}}{H_{\text{ref}}}}. \quad (1.15)$$

Scaling both pressure and extra-stress by the viscous stress unit $\eta_0 U_{\text{ref}}/H_{\text{ref}}$ ensures that all stress and pressure terms remain $\mathcal{O}(1)$ in low-Reynolds regimes, eliminating singular perturbation behavior ($1/Re \rightarrow \infty$) as $Re \rightarrow 0$.

Dropping asterisks for notational simplicity, the dimensionless system is gov-

erned by the following parameters:

$$Re_{\text{scale}} = \frac{\rho U_{\text{ref}} H_{\text{ref}}}{\eta_0} \quad (\text{Scaling Reynolds number}), \quad (1.16)$$

$$Wi = \frac{\lambda U_{\text{ref}}}{H_{\text{ref}}} \quad (\text{Weissenberg number}), \quad (1.17)$$

$$\mu_s^* = \frac{\mu_s}{\eta_0} \quad (\text{Dimensionless solvent viscosity}), \quad (1.18)$$

$$\mu_p^* = \frac{\mu_p}{\eta_0} \quad (\text{Dimensionless polymeric viscosity}). \quad (1.19)$$

From these parameters, the physical viscosity ratio $\beta = \mu_s/(\mu_s + \mu_p) = \mu_s^*/(\mu_s^* + \mu_p^*)$ and the physical Reynolds number $Re_{\text{phys}} = \rho U_{\text{ref}} H_{\text{ref}}/(\mu_s + \mu_p) = Re_{\text{scale}}/(\mu_s^* + \mu_p^*)$ can be uniquely determined.

1.4.2 Dimensionless Vector System

The non-dimensional governing equations in steady 2D flow become:

$$\nabla \cdot \mathbf{u} = 0, \quad (1.20)$$

$$Re_{\text{scale}}(\mathbf{u} \cdot \nabla \mathbf{u}) + \nabla p - \mu_s^* \nabla^2 \mathbf{u} - \nabla \cdot \boldsymbol{\tau} = \mathbf{0}, \quad (1.21)$$

$$Wi(\mathbf{u} \cdot \nabla \boldsymbol{\tau} - (\nabla \mathbf{u})^T \cdot \boldsymbol{\tau} - \boldsymbol{\tau} \cdot \nabla \mathbf{u}) + \boldsymbol{\tau} - 2\mu_p^* \mathbf{D} = \mathbf{0}. \quad (1.22)$$

In this formulation, the constitutive equation (1.22) depends strictly on the rheological parameters (Wi, μ_p^*) and is decoupled from inertia (Re_{scale}) and solvent viscosity (μ_s^*), which is crucial for staged parameter identifiability in inverse PINN solvers.

1.4.3 Explicit 2D Cartesian PDE Residuals

Expanding Equations (1.21) and (1.22) in 2D Cartesian coordinates (x, y) with $\mathbf{u} = (u, v)^T$ and $\boldsymbol{\tau} = \begin{pmatrix} \tau_{xx} & \tau_{xy} \\ \tau_{xy} & \tau_{yy} \end{pmatrix}$ yields five scalar PDE residuals. Symmetry

of the stress tensor ($\tau_{xy} = \tau_{yx}$) reduces the system to two momentum residuals (f_u, f_v) and three constitutive residuals ($f_{\tau_{xx}}, f_{\tau_{yy}}, f_{\tau_{xy}}$):

$$f_u = Re_{\text{scale}} \left(u \frac{\partial u}{\partial x} + v \frac{\partial u}{\partial y} \right) + \frac{\partial p}{\partial x} - \mu_s^* \left(\frac{\partial^2 u}{\partial x^2} + \frac{\partial^2 u}{\partial y^2} \right) - \left(\frac{\partial \tau_{xx}}{\partial x} + \frac{\partial \tau_{xy}}{\partial y} \right) = 0, \quad (1.23)$$

$$f_v = Re_{\text{scale}} \left(u \frac{\partial v}{\partial x} + v \frac{\partial v}{\partial y} \right) + \frac{\partial p}{\partial y} - \mu_s^* \left(\frac{\partial^2 v}{\partial x^2} + \frac{\partial^2 v}{\partial y^2} \right) - \left(\frac{\partial \tau_{xy}}{\partial x} + \frac{\partial \tau_{yy}}{\partial y} \right) = 0, \quad (1.24)$$

$$f_{\tau_{xx}} = \tau_{xx} + Wi \left(u \frac{\partial \tau_{xx}}{\partial x} + v \frac{\partial \tau_{xx}}{\partial y} - 2 \frac{\partial u}{\partial x} \tau_{xx} - 2 \frac{\partial u}{\partial y} \tau_{xy} \right) - 2 \mu_p^* \frac{\partial u}{\partial x} = 0, \quad (1.25)$$

$$f_{\tau_{yy}} = \tau_{yy} + Wi \left(u \frac{\partial \tau_{yy}}{\partial x} + v \frac{\partial \tau_{yy}}{\partial y} - 2 \frac{\partial v}{\partial x} \tau_{xy} - 2 \frac{\partial v}{\partial y} \tau_{yy} \right) - 2 \mu_p^* \frac{\partial v}{\partial y} = 0, \quad (1.26)$$

$$f_{\tau_{xy}} = \tau_{xy} + Wi \left(u \frac{\partial \tau_{xy}}{\partial x} + v \frac{\partial \tau_{xy}}{\partial y} - \frac{\partial u}{\partial x} \tau_{xy} - \frac{\partial u}{\partial y} \tau_{yy} - \frac{\partial v}{\partial x} \tau_{xx} - \frac{\partial v}{\partial y} \tau_{xy} \right) - \mu_p^* \left(\frac{\partial u}{\partial y} + \frac{\partial v}{\partial x} \right) = 0. \quad (1.27)$$

1.5 Four-Roll Mill Boundary Conditions

The four-roll mill is a canonical experimental and computational benchmark designed to generate controlled two-dimensional planar extensional flow with a central stagnation point [5, 6]. Originally devised by Taylor [5] to study droplet deformation and breakup under well-defined extensional kinematics, the apparatus has since become a standard test case for validating viscoelastic flow solvers.

As illustrated in Figure 1.1, the setup consists of four cylindrical rollers placed symmetrically inside a square domain.

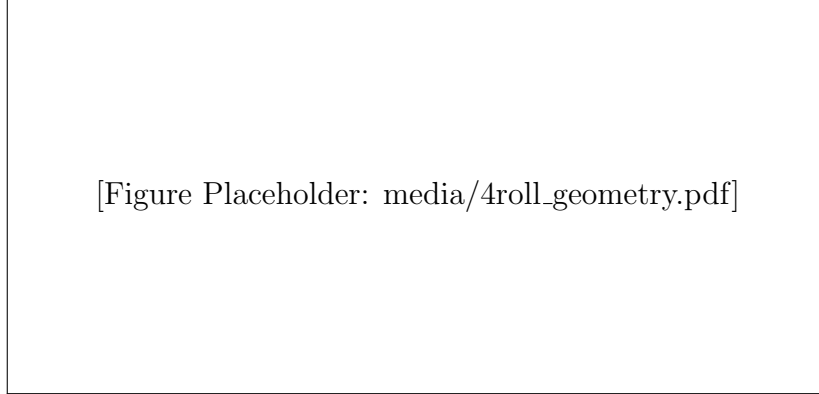


Figure 1.1: Schematic representation of the four-roll mill geometry, indicating roller counter-rotation speeds $\boldsymbol{\Omega}_k$, the central hyperbolic stagnation point $(0, 0)$, and inflow/outflow axes.

1.5.1 Geometry and Roller Rotation Dynamics

The domain consists of a square cavity containing four identical cylindrical rollers of radius R positioned symmetrically. The rollers rotate at equal angular speeds Ω . Adjacent rollers rotate in opposite directions:

Top-Left roller (TL): rotates clockwise ($\boldsymbol{\Omega}_{\text{TL}} = -\Omega \mathbf{e}_z$).

Top-Right roller (TR): rotates counter-clockwise ($\boldsymbol{\Omega}_{\text{TR}} = +\Omega \mathbf{e}_z$).

Bottom-Left roller (BL): rotates counter-clockwise ($\boldsymbol{\Omega}_{\text{BL}} = +\Omega \mathbf{e}_z$).

Bottom-Right roller (BR): rotates clockwise ($\boldsymbol{\Omega}_{\text{BR}} = -\Omega \mathbf{e}_z$).

This counter-rotating configuration draws fluid inward along the vertical axis (compression axis) and expels it outward along the horizontal axis (extension axis), creating a *hyperbolic* (saddle-type) stagnation point at the origin $(0, 0)$. At this point, the velocity vanishes ($\mathbf{u} = \mathbf{0}$) while the velocity gradient tensor $\nabla \mathbf{u}$ possesses real eigenvalues of equal magnitude and opposite signs, corresponding to

simultaneous extension and compression along orthogonal axes. Near the origin, the velocity field approaches pure planar extension:

$$u(x, y) = \dot{\epsilon} x, \quad v(x, y) = -\dot{\epsilon} y, \quad (1.28)$$

where $\dot{\epsilon} \propto \Omega$ is the extensional strain rate. This kinematic configuration makes the four-roll mill particularly relevant for viscoelastic flows, as the persistent extensional deformation near the stagnation point can stretch polymer chains far from their equilibrium configuration, amplifying elastic stress effects.

1.5.2 Boundary Conditions for Velocity and Stress

To complete the boundary value problem in CFD, appropriate boundary conditions are imposed:

1. **No-Slip Velocity BCs on Roller Walls (Γ_{rolls}):** On the surface of each roller $k \in \{\text{TL, TR, BL, BR}\}$, the fluid velocity matches the tangential roller wall speed:

$$\mathbf{u}|_{\Gamma_k} = \boldsymbol{\Omega}_k \times (\mathbf{x} - \mathbf{x}_k), \quad (1.29)$$

where $\mathbf{x}_k$ is the center coordinate of roller k .

2. **Outer Boundary Conditions (Γ_{outer}):** On the outer bounding walls of the cavity, no-slip velocity conditions ($\mathbf{u} = \mathbf{0}$) or prescribed inflow/outflow velocity profiles corresponding to the extensional field (1.28) are specified.
3. **Polymeric Extra-Stress Wall BCs:** For viscoelastic fluids, stress boundary conditions on solid walls must be consistent with the local kinematics. On the rotating rollers, the stress tensor $\boldsymbol{\tau}$ satisfies either the local steady-state constitutive relation evaluated at the wall velocity gradient $\nabla \mathbf{u}|_{\Gamma_{\text{rolls}}}$ or

is constrained via Dirichlet conditions extracted from COMSOL reference simulations.

1.5.3 Pressure Pinning and Gauge Fix in Enclosed Domains

In incompressible fluid mechanics, the pressure p appears in the momentum equation solely through its spatial gradient ∇p . As a consequence:

If a velocity boundary condition (Dirichlet no-slip) is specified along the entire closed boundary $\partial\Omega = \Gamma_{\text{rolls}} \cup \Gamma_{\text{outer}}$, the pressure field is determined only up to an arbitrary additive constant C :

$$p(\mathbf{x}) \rightarrow p(\mathbf{x}) + C. \quad (1.30)$$

Without a pressure reference point, the continuous Neumann problem for pressure lacks a unique solution, causing numerical solvers (both traditional FEM/FVM and PINN optimization) to suffer from severe pressure drift.

To ensure mathematical well-posedness and unique pressure recovery, a single-point pressure pin (gauge condition) is enforced at a reference location $\mathbf{x}_0$ (typically the stagnation point or a domain corner):

$$p(\mathbf{x}_0) = p_0 = 0. \quad (1.31)$$

Fixing $p(\mathbf{x}_0) = 0$ removes the hydrostatic gauge ambiguity, allowing the numerical solver to uniquely reconstruct the absolute pressure field across the four-roll mill domain. In the PINN framework, this gauge condition is enforced as an additional penalty term in the composite loss function, $L_{\text{pin}} = |\hat{p}(\mathbf{x}_0) - p_0|^2$, which anchors the predicted pressure to the reference value during optimization.

Chapter 2

Physics-Informed Neural Networks

In this chapter, we present a literature review of Physics-Informed Neural Networks (PINNs) and their application to complex fluid mechanics. The discussion is structured into three main sections: first, a general overview of the PINN paradigm and its advantages over classical numerical methods and purely data-driven models; second, an in-depth analysis of the foundational framework introduced by Raissi et al. [7]; and third, a detailed examination of the Viscoelastic-Net framework developed by Thakur, Raissi, and Ardekani [8], which represents a significant methodological advance for stress discovery and constitutive model identification in viscoelastic fluid dynamics.

2.1 General Overview of Physics-Informed Neural Networks

Over the past decade, scientific machine learning (SciML) has emerged as a transformative discipline at the intersection of deep learning and mathemati-

cal physics [9]. Traditional numerical techniques for solving partial differential equations (PDEs)—such as the Finite Element Method (FEM), Finite Volume Method (FVM), and Finite Difference Method (FDM)—rely on spatial discretization grids. While highly accurate, these conventional solvers face significant limitations:

Meshing Complexity: Generating high-quality computational meshes around intricate geometries or singular flow regions requires substantial computational overhead.

Curse of Dimensionality: Grid-based methods scale poorly to high-dimensional state spaces or parametric optimization problems.

Inability to Handle Sparse Data: Traditional CFD solvers are strictly designed for forward initial-boundary value problems and cannot naturally ingest sparse experimental measurements to solve inverse problems.

Conversely, standard black-box machine learning models—such as convolutional neural networks (CNNs) for spatial field reconstruction, recurrent architectures for temporal dynamics, and generative adversarial networks (GANs) for stochastic flow synthesis—require vast quantities of high-fidelity training data. Furthermore, because they lack physical awareness, these purely data-driven networks often produce predictions that violate basic conservation laws (such as mass or momentum conservation) when evaluated outside their training distribution, and they offer no guarantee of physical consistency in extrapolation regimes.

Physics-Informed Neural Networks (PINNs) bridge this gap by leveraging the universal approximation property of neural networks [10] while embedding the governing differential equations directly into the training process [7]. By penalizing violations of physical laws during optimization, PINNs act as physics-constrained surrogate models that offer several distinct advantages over both

traditional solvers and black-box approaches:

Mesh-free operation: PINNs evaluate the solution at arbitrary collocation points without requiring a spatial discretization grid, eliminating meshing overhead.

Natural inverse problem support: Unknown physical parameters can be jointly optimized alongside the network weights, enabling simultaneous field reconstruction and parameter identification.

Sparse and noisy data assimilation: PINNs can incorporate scattered experimental measurements, using the governing equations as a regularizer to fill in spatial and temporal gaps.

Continuous differentiable solutions: The neural network output provides an infinitely differentiable analytical representation of the solution field throughout the entire domain.

2.2 The Foundational PINN Framework: Raissi et al. (2019)

The foundational methodology of PINNs was formally introduced by Raissi, Perdikaris, and Karniadakis in their landmark 2019 publication [7]. Their key innovation was to leverage *Automatic Differentiation* (AD) to seamlessly integrate general non-linear PDEs into deep learning computational graphs.

2.2.1 Automatic Differentiation via Computational Graphs

Consider a general non-linear partial differential equation governing a continuous state variable $u(\mathbf{x}, t)$:

$$\mathcal{F}[u(\mathbf{x}, t); \boldsymbol{\lambda}] = 0, \quad \mathbf{x} \in \Omega, \quad t \in [0, T], \quad (2.1)$$

where $\mathcal{F}$ is a differential operator and $\boldsymbol{\lambda}$ denotes a set of physical parameters.

In the PINN framework, the unknown solution $u(\mathbf{x}, t)$ is approximated by a deep Multi-Layer Perceptron (MLP) parameterized by trainable weights and biases $\boldsymbol{\theta}$:

$$\hat{u}(\mathbf{x}, t) = \mathcal{N}_{\boldsymbol{\theta}}(\mathbf{x}, t). \quad (2.2)$$

Specifically, this MLP architecture consists of an input layer for the coordinates, hidden layers equipped with non-linear activation functions (typically `tanh` or `Swish`), and an output layer for the predicted field variables. The parameters are commonly initialized using the Xavier/Glorot scheme.

To bridge this representation with the underlying physics, analytical derivatives are required. Instead of approximating spatial and temporal derivatives using finite-difference stencils on a grid, PINNs compute exact analytical derivatives using Automatic Differentiation (AD) via reverse-mode accumulation in computational graphs (e.g., PyTorch’s `autograd`). This approach is fundamentally superior for PINNs because AD provides exact derivatives (up to floating-point precision) without introducing discretization error or requiring an underlying grid. Derivatives such as $\partial \hat{u} / \partial x$ or $\partial^2 \hat{u} / \partial x^2$ are evaluated to machine precision by applying the chain rule directly across the hidden layers of the network.

2.2.2 Physics-Constrained Loss Function Formulation

To enforce the governing PDE (2.1), a *PDE residual network* $f(\mathbf{x}, t)$ is constructed by substituting the neural network output $\hat{u}$ into the differential operator:

$$f(\mathbf{x}, t) \equiv \mathcal{F}[\mathcal{N}_{\boldsymbol{\theta}}(\mathbf{x}, t); \boldsymbol{\lambda}]. \quad (2.3)$$

The network parameters $\boldsymbol{\theta}$ are optimized by minimizing a composite loss function $L(\boldsymbol{\theta})$ defined over three distinct sets of points [7]:

$$L(\boldsymbol{\theta}) = w_{\text{data}} L_{\text{data}}(\boldsymbol{\theta}) + w_{\text{bc}} L_{\text{bc}}(\boldsymbol{\theta}) + w_{\text{pde}} L_{\text{pde}}(\boldsymbol{\theta}), \quad (2.4)$$

where:

$$L_{\text{data}}(\boldsymbol{\theta}) = \frac{1}{N_{\text{data}}} \sum_{i=1}^{N_{\text{data}}} |\mathcal{N}_{\boldsymbol{\theta}}(\mathbf{x}_i^{\text{data}}, t_i^{\text{data}}) - u_i^{\text{data}}|^2, \quad (2.5)$$

$$L_{\text{bc}}(\boldsymbol{\theta}) = \frac{1}{N_{\text{bc}}} \sum_{j=1}^{N_{\text{bc}}} |\mathcal{B}[\mathcal{N}_{\boldsymbol{\theta}}(\mathbf{x}_j^{\text{bc}}, t_j^{\text{bc}})] - g_j^{\text{bc}}|^2, \quad (2.6)$$

$$L_{\text{pde}}(\boldsymbol{\theta}) = \frac{1}{N_{\text{pde}}} \sum_{k=1}^{N_{\text{pde}}} |f(\mathbf{x}_k^{\text{pde}}, t_k^{\text{pde}})|^2. \quad (2.7)$$

Here, $\mathcal{B}$ represents boundary operators, while N_{data} , N_{bc} , and N_{pde} denote the number of data observation points, boundary condition points, and interior PDE collocation points, respectively. Physically, L_{data} fits the model to empirical observations, L_{bc} enforces the requisite boundary and initial conditions, and L_{pde} enforces the governing physical laws. The weights w_{data} , w_{bc} , and w_{pde} control the relative importance of these constraints; their tuning is a recognized challenge due to gradient pathologies [11].

Transitioning to practical applications, this unified loss minimization paradigm

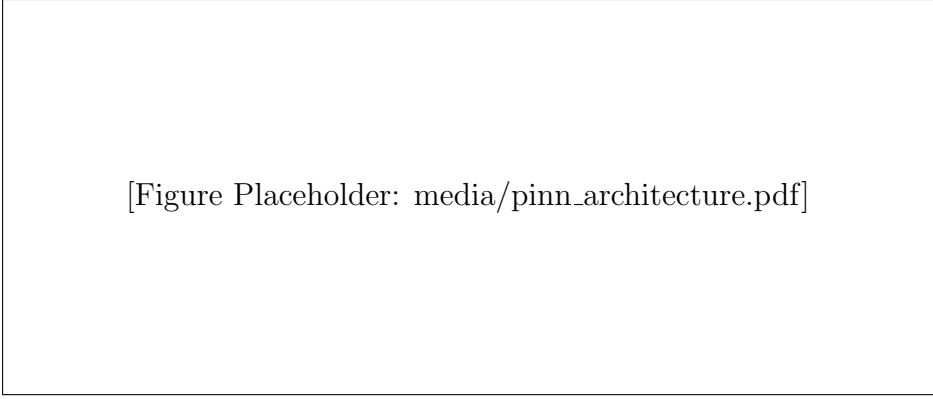
naturally supports two distinct scientific computing modes:

1. **Forward Problems (Data-Driven Solutions):** When the physical parameters λ are fully known, PINNs optimize θ to solve for the continuous field variables throughout the spatio-temporal domain without requiring a grid.
2. **Inverse Problems (Data-Driven Discovery):** When the underlying parameters λ are unknown (e.g., uncharacterized fluid viscosity or relaxation time), λ is treated as a set of extra trainable parameters alongside θ . By training on a small set of sparse field observations, PINNs simultaneously reconstruct the continuous state fields and identify the exact physical parameters λ^* .

Despite these broad capabilities, PINN training introduces specific challenges: spectral bias (the network’s tendency to learn low-frequency solutions first), gradient imbalance between competing loss terms, and a strong sensitivity to the spatiotemporal distribution of collocation points [11].

2.3 ViscoelasticNet: Stress Discovery and Model Selection

Building upon the foundational PINN methodology, Thakur, Raissi, and Ardekani [8] developed *ViscoelasticNet*, a specialized physics-informed neural network framework published in the *Journal of Non-Newtonian Fluid Mechanics*. Their work represents a major breakthrough in complex fluid mechanics, addressing long-standing experimental and computational bottlenecks in viscoelastic flow modeling.



[Figure Placeholder: media/pinn_architecture.pdf]

Figure 2.1: Conceptual diagram of the ViscoelasticNet framework (Thakur et al., 2024), illustrating the physics-informed architecture for learning hidden stress tensors and identifying viscoelastic constitutive models using velocity field inputs. The diagram highlights the input layer (spatiotemporal coordinates), hidden layers, output layer (predicted fields), and the physics loss evaluation.

2.3.1 The Challenge of Viscoelastic Flow Characterization

Viscoelastic fluids—such as polymer solutions, biological fluids, and molten plastics—exhibit complex non-linear behaviors, including shear-thinning, stress relaxation, and elastic memory [1]. Modeling these fluids presents two major challenges:

Experimental Measurement Limits: While modern optical techniques (such as Particle Image Velocimetry - PIV and flow birefringence [12]) can measure velocity fields $\mathbf{u} = (u, v)$ with high spatial resolution, direct non-intrusive experimental measurement of internal stress tensor fields $\boldsymbol{\tau} = (\tau_{xx}, \tau_{xy}, \tau_{yy})$ and pressure p inside complex flows remains extremely difficult or impossible.

Computational Solvers Instability: Traditional computational viscoelasticity solvers (FEM/FVM) suffer from the *High Weissenberg Number Problem* (HWNP). At elevated elastic rates ($Wi > 0.5$), numerical discretization of the upper-convected stress advection term $(\mathbf{u} \cdot \nabla \boldsymbol{\tau})$ leads to severe numer-

ical instabilities, artificial stress singularities, and catastrophic solver divergence [4]. Numerically, the upper-convected derivative introduces a hyperbolic character to the constitutive equation. Traditional methods struggle because the stress equation transitions from elliptic to hyperbolic behavior at high Weissenberg numbers. Stabilization techniques used in FEM, such as Streamline Upwind Petrov-Galerkin (SUPG) and the log-conformation transformation, can mitigate these issues but often fail at sufficiently high elasticity or introduce excessive numerical diffusion.

Given these limitations, deep learning provides a compelling alternative methodology.

2.3.2 Why ViscoelasticNet is Revolutionary

ViscoelasticNet [8] solves these challenges by reformulating viscoelastic stress characterization as a physics-informed inverse discovery problem. The revolutionary contributions of the ViscoelasticNet framework can be summarized as follows:

1. Latent Stress and Pressure Discovery from Velocity Data Only:

ViscoelasticNet requires *only velocity field observations* (which can be readily obtained from experimental PIV or synthetic datasets) as input. By embedding the momentum conservation and Oldroyd-B constitutive equations into the loss function, the network infers the hidden, unmeasurable polymeric extra-stress tensor $\boldsymbol{\tau}$ and pressure field p throughout the entire domain without needing any experimental stress measurements. More precisely, ViscoelasticNet uses the momentum equation $\nabla \cdot \boldsymbol{\sigma} = \rho(\mathbf{u} \cdot \nabla \mathbf{u})$ to constrain the stress divergence, while the constitutive equation constrains the full stress tensor field. The combination of these physics constraints with empirical velocity data creates a well-posed inverse problem.

2. **Robustness to High Weissenberg Numbers:** By framing PDE solving as an optimization task over continuous neural network spaces rather than step-by-step time integration on a spatial mesh, ViscoelasticNet bypasses the mesh-dependent numerical instabilities associated with the HWNP [8].
3. **Constitutive Model Selection and Parameter Identification:** ViscoelasticNet provides a data-driven mechanism to select the correct constitutive model (such as Oldroyd-B, linear PTT, or Giesekus) for an uncharacterized fluid. By optimizing residual losses corresponding to candidate constitutive models, the framework identifies both the governing constitutive model and its underlying physical parameters (relaxation time λ , polymeric viscosity μ_p , non-linear mobility parameters).

To validate this approach, Thakur et al. [8] evaluated the framework on specific benchmark flows: steady Couette flow, flow past a circular cylinder, and a four-roll mill. The four-roll mill configuration proved particularly challenging due to the strong extensional kinematics near the stagnation point, which rigorously tests the network’s capacity to resolve steep stress gradients.

By demonstrating that continuous stress fields and constitutive parameters can be discovered non-intrusively from velocity data alone, Thakur et al. [8] established a mathematically rigorous foundation for applying physics-informed machine learning to complex rheological flows.

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